INDEPENDENT PROTOTYPE · FICTIONAL PATIENTS ONLY · FOR FEEDBACK

Teach-Back Companion and Instruction Fidelity Evaluator

EHR assistants now explain records to patients. This prototype targets the next two questions: **did the patient understand?** and **does the simplified or translated instruction still say the same thing?**

Teach-back (patient side)

- Each discharge instruction is shown verbatim beside a plain version.
- The patient explains it in their own words. The tool shows what they covered, what is missing, and possible misunderstandings ("I can stop the blood thinner when I feel better"), always quoting the original and pointing to the care team.
- Emergency, crisis and medication-change messages are screened first and sent to human help.

Fidelity evaluator (care-team side)

- Checks any rewrite (human, AI or translated) against the original for doses, frequencies, tablet counts, limits, "do not" statements, 911 instructions, medicine names and reading level.
- Benchmark: on 100 independently written rewrites, flagged 97% of dangerous changes; 90% of faithful rewrites passed (before tuning).

Why it may matter

- Teach-back is recommended (AHRQ Health Literacy Toolkit, Tool 5) but time-consuming.
- A 2026 randomised trial in internal medicine found physician-reviewed, AI-generated patient discharge summaries increased patient activation (Rust et al., *Lancet Digit Health* 2026;8(6):100991). The fidelity evaluator is a candidate aid for that review step.

Status and boundaries

- Measured on 390 cases written by an independent reviewer, before tuning: flagged 88% of dangerous teach-back explanations with no false alarms; recognised 87% of emergencies and 73% of crisis messages. **Not a safety net**; an aid for clinician-led teach-back.
- 247 automated tests; browser and WCAG automated checks pass; five review and measurement rounds.
- Fictional patients; draft education text not yet clinician-reviewed; rule-based, not clinically validated.
- No EHR connection, no PHI, nothing typed is stored.

Try it (15 min): open the repository in GitHub Codespaces; the demo opens in the browser. Or `pip install -e . && teachback serve`.
Ask: 20 minutes of feedback on whether this problem is worth solving at UW Medicine. This is not a pilot request.
Code: https://github.com/fffn259m2n-cell/UWMC/tree/main/teach-back-companion

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